Costs 'C' to divide at each step.

Date

Recurrence relation of Binary Search = $\log_2(n) * C$

LECTURE-4

Monday 3/6/26.

Factorial:

(n-b) instead of (n/b)

$T(n) = T(n-1) + C_1$ → checks if base case is hit in each iteration.

$T(n) \xrightarrow{C_1} T(n-1) \xrightarrow{C_1} T(n-2) \cdots\cdots T(2) \xrightarrow{C_2} T(1)$ → Terminal Nodes

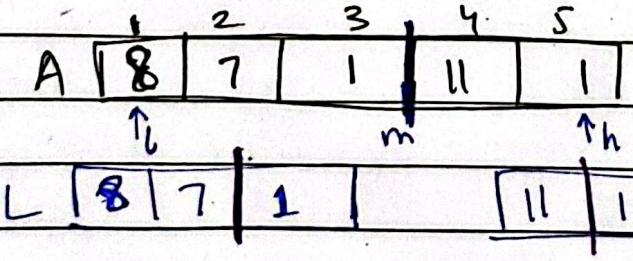
* Workdone at each level will be added for total work

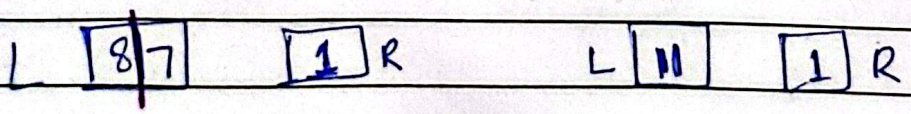
$T(n) = (n-1) \cdot C_1 + C_2$

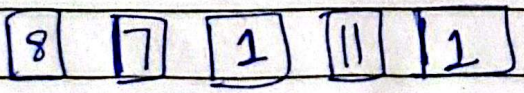
$\therefore T(n) = \boxed{\Theta(n)}$. as constants are cancelled

Merge Sort (Divide & Conquer):

Keep breaking the array from middle until we've single element in each section which makes it sorted (as single element)



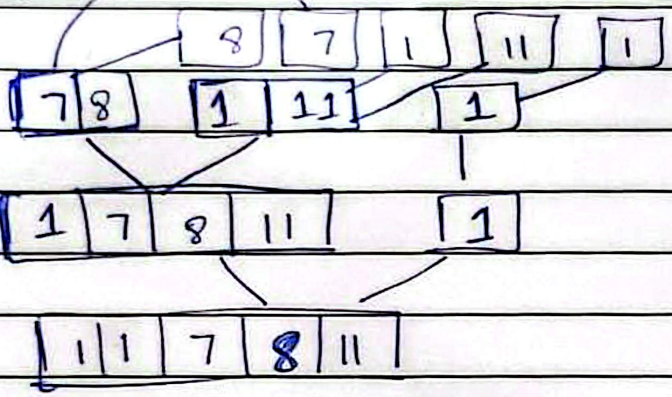




* we'll stop when l==h in each section.

Merging Routine

Date _____



for this we'll be needing nL and nR (sizes of left & right) as we need to create the actual right & left arrays.

So ; $nL = m - l + 1$

$nR = h - m$.

+ Now create L & R arrays.

```
Merge Sort (A [1:n], l, h) {
    if (l ≥ h) return
    else {
        m = (l+h) / 2
        MergeSort ( A , l, m)
        Merge Sort (A , m+1 , h)
        Merge Routine (A, l, m, h) // T(n) for merge routine is O(n)
    }                                  as we're three single loops
}
```